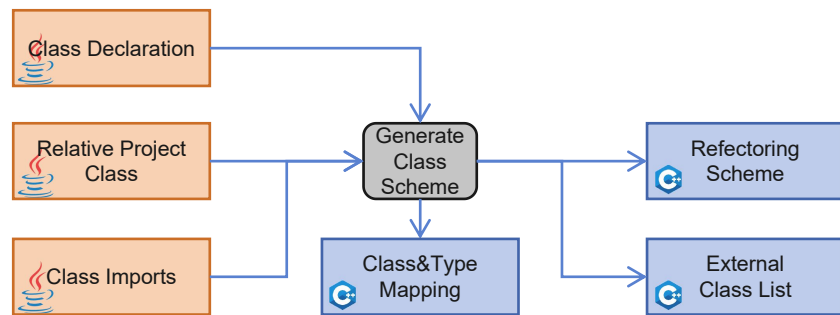




Following is a java file information from the project, now it is supposed to be translated to a C++ header Including declarations of all fields and methods.

`{java_class declaration, java imports}`

Carefully read the java code, then specify a detailed translation schema, including the following contents:

1. The necessary refactoring to align differences in language features, such as: use a pure virtual base class to replace an interface, use RTTI to replace Java's instanceof operator, add extra fields and methods to imitate reflection, etc..;
2. Any external C++ classes(include enum class, virtual class, etc.) not in the current project and not in standard library need to be implemented in the future;
3. Mapping of data types and classes. The mapped class may be a primitive C++ type, composite type, a class in the C++ Standard Library or a custom class that needs to be implemented;
4. Other additional explanations (if any).

The available headers in the current project are as follows. If you need to use a header that is not included in them, please add it to the list of external classes.

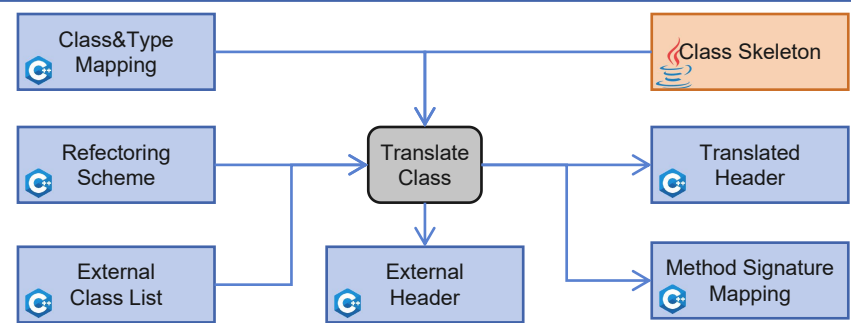
`{relative project classes}`

The answer should be in the format of json:

`{json_example}`

Notes:

1. All the above require specific solutions to be provided, DO NOT provide ambiguous answers, and DO NOT directly provide C++ code.
2. For classes in the Java standard library, try to use corresponding classes in the C++ standard library. If there is no corresponding class, add a external class instead.



The following is a part of java file from the source repository. it contains the member variables and member functions(the function implements are omitted). Now it is supposed to be translated to a C++ header file.

`{java_skeleton}`

The following is the translation scheme:

`{scheme}`

According to the above translation plan, provide the translated result, your answer should include:

1. The translated code;
2. Mapping between translated fields/methods and original java fields/methods.
3. The necessary external class declarations in translation plan(if any), they will be added to the translated repository as additional header files.

The available header files in the current project are as follows. If you need to use a header file that is not included in them, please add it to the list of external header files.

`{header_file_list}`

The answer should be in the format of json:

`{json_example}`

Notes:

1. Only header file is required, don't implement the methods;
2. the "method_mapping" should at least include the following method signatures:

`{all_method_signatures}`